



# Solving a real-world nurse rostering problem by Simulated Annealing

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## ABSTRACT

Designing high quality nurse rostering plans is essential for health care facilities in order to guarantee efficiency, safety and quality-of-care balanced with staff well-being.

We introduce a new real-world formulation for the nurse rostering problem, arising in many Italian healthcare institutions, which has been developed in collaboration with a primary software company in the field. It considers nurses with different skills, special shifts depending on the skills, time work-load limits, and different types of days-off. In addition, preferences and incompatibilities between nurses are taken into account.

We propose a MIP model and a local search method, driven by a Simulated Annealing metaheuristic, based on a combination of two neighborhoods. The solution method was tested on 34 real-world instances coming from various healthcare institutions in North Italy. The dataset is available at <https://bitbucket.org/satt/nrp-instances>, along with our best solutions.

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## 1. Introduction

Nurse rostering (NR) is one of the crucial problems that every hospital has to manage periodically. Within the extensive literature on staff scheduling in general [1], many papers have been devoted to nurse rostering problems [2]. Different formulations of the NR problem have been studied; among the others, those presented for the two *International Nurse Rostering Competitions*, INRC-I [3] and INRC-II [4], have achieved considerable interest in the scientific community, due to the contextual competitions. However, as recently stated by Petrovic [5], there still exists a gap between personnel scheduling research and implementation in practice, which can be reduced by engaging more with schedulers and software developers.

In this research project, practitioners and researchers collaborated to address a NR problem in the context of Italian institutions, and developed an effective solution approach that constitutes the algorithmic core of an industrial application, completed with a graphical front-end, and integrated with the existing HR software infrastructure.

To this aim, a novel real-world formulation for NR is introduced, which is quite different from the ones already proposed

in the literature, given that it pursues accurately to the terms of the Italian contract of employment for nurses. The problem is proposed here in its full complexity, with no concession to judicious simplifications.

For example, it prescribes that there must be a minimum break of 11 h between two consecutive shifts. In addition, the novel notion of “rest-day” is introduced, which is the day immediately after a night shift, such that the only permitted succession involving the end of a sequence of night shifts is *night/rest-day/day-off*.

For this problem, we propose a MIP model and implement a local search method based on the Simulated Annealing metaheuristic, building upon a previous work [6] in which such a technique was successfully applied to the simpler problem of the INRC-II.

In a reasonable computational time, the local search method has been able to compute good quality solutions for all instances, in which all hard constraints are satisfied and most of the soft ones have zero cost or a very small values.

As an important byproduct of this work, we collected a set of 34 real-world instances, coming from different health facilities in northern Italy, which, properly anonymized, are available to the community for future comparisons.

The remainder of the paper is organized as follows. Section 2 defines our NR problem whose mathematical model is provided in Section 3. In Section 4, we review the literature about real-world NR problems. Section 5 presents the solution method, and computational results are shown and discussed in Section 6.

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Finally, Section 7 draws some conclusions and identifies lines for future research. The system infrastructure that embeds the solver is described in [Appendix](#).

## 2. Problem definition

In the real-world NR problem tackled, the goal is to create a rostering plan of a health facility for a fixed planning horizon. In detail, the most important notions defining the problem are the following ones:

**Shifts & Periods:** A shift is a nurse's scheduled time of work; it has a start time and a duration and it belongs to a period. Periods are specific parts of the day time, such that the minimum and maximum number of consecutive shifts permitted is aggregated at the level of the same period.

**Absences:** There are various types of non working days for a nurse, and they have a different impact on the constraints as explained below. The *day-off* is when the nurse is normally scheduled to be off work. The *rest-day* is the specific day-off after the last night shift in a night sequence. The *holiday* is when the nurse is voluntarily taking a vacation day. The *unavailability* is when the nurse is working on another ward or doing another duty.

**Skills:** A skill identifies a professional profile or a service. Skills are related to shifts so that each skill is associated to a list of feasible shifts. A nurse on duty with that skill can be assigned only to a shift associated to the skill.

**Contracts:** Each nurse works under a specific employment contract. The contract specifies for the planning period: the minimum and the maximum number of working days, the minimum and maximum working time (in minutes), the minimum number of consecutive days off and the required number of preceding working days, the maximum number of consecutive days off, the maximum number of consecutive working days, the maximum number of working Sundays, weekends and public holidays. In addition, it defines the minimum distance (in days) between two sequences of night shifts, the working days in a week (e.g., from Monday to Friday), the average daily working time (in minutes), and the minimum number of rest hours after a given number of consecutive shifts.

**Nurses:** Nurses are characterized by their gender, their skills, the set of feasible shifts, their standard skill and shift, and the sequence of the last assigned shifts in the period preceding the horizon plan (history data).

**Pre-assignments, Period Off Requests, & Preferences:** It is possible to define a pre-assignment, working day or period off request specifying the nurse, the day, the shift and the skill (or a day-off), and a priority level (among *fixed*, *high*, *medium*, *low*). In case of request of period-off for a day, any shift of the given period should be avoided. A pre-assignment (or request) whose level is different than *fixed* is considered as a preference.

**Workteams and Incompatibilities:** Some nurses can be grouped into workteams, which collect favorite colleagues, whereas incompatibilities are expressed for pair of nurses. Workteams and incompatibilities between nurses have different levels (*high*, *medium*, *low*).

**Coverage:** For each day of the planning horizon, for each skill and for each period, the minimum, optimal and maximum coverage is expressed in terms of number of required nurses.

The periods considered in our cases are four, namely *Early*, *Daily*, *Late*, and *Night*, plus a special long one called *Day*. For the period *Day*, no specific coverage request is expressed, as a shift of this period covers both an *Early* and a *Late* periods.

The problem consists in assigning to each nurse, for each day, a shift and a skill or a non working status (day-off, rest-day, holiday, or unavailability), satisfying the hard and soft constraints described below.

Our problem involves a large set of constraints. In order to make this set manageable and understandable, we decided to classify them based on who is the most prominent interest bearer for it. In detail, they are divided into three categories: A (for administration), N (for nurses), and P (for patients), depending on which entity is most interested in the satisfaction of the specific constraint.

### Administration oriented constraints.

**A1. Understaffing:** For each day, for each skill and for each period, the number of assigned nurses must be greater or equal to minimum coverage requirements. Violations of this constraint for *Night* period are more penalized.

**A2. Exceeding Staffing:** For each day, for each skill and for each period, the number of assigned nurses must not exceed the maximum coverage requirements.

**A3. Nurse/Skill/Shift Compatibility:** Every shift refers to one or more skills and it can be assigned to a nurse only if the nurse is suitable for that skill and their employment contract provides for that shift.

**A4. Single Assignment:** Each day, only a single shift can be assigned to a nurse.

**A5. Total Assignments:** For each nurse, the contract establishes the minimum and the maximum working days in the planning horizon. A holiday counts as a working day and it contributes to the total assignments counter, whereas an unavailability does not.

**A6. Total Working Time:** For each nurse, the contract sets the minimum and the maximum working time (in minutes) during the planning horizon. For each holiday, the average daily working time defined in the contract is added to the total working time counter.

### Nurse oriented constraints.

**N1. Complete Weekend:** A nurse should work on both Saturday and Sunday or neither.

**N2. Consecutiveness:** It is defined the minimum and the maximum number of consecutive working days, days-off and shifts of a particular period that can be assigned to a nurse in the planning horizon. Consecutiveness constraints with respect to the previous planning period are managed by storing suitable history data.

**N3. Fixed Requests:** A request of pre-assignment, holiday, working day or period-off with level *fixed* must be respected.

**N4. Incompatibility:** Incompatibility between nurses working in the same period should be avoided, with different levels of priority (*high*, *medium*, *low*). If two incompatible nurses work in the same period, this is penalized proportionally to their incompatibility level.

- N5. Total Weekends:** For each nurse, the contract fixes the maximum working weekends in the planning horizon. If the nurse works on Saturday and/or on Sunday, it is considered a “working” weekend.
- N6. Max Sunday Assignments:** For each nurse, the contract limits the maximum number of Sunday as working day in the planning horizon.
- N7. Max Working Holidays:** For each nurse, the contract limits the maximum number of public holidays as working day in the planning horizon.
- N8. Minimum Days Off After Consecutive Assignments:**  
For each nurse, the contract sets a minimum number of days off that should follow a given number of consecutive working days (e.g., one day off after six consecutive assignments to any shift).
- N9. Minimum Distance Between Nights:** Two sequences of night shifts should be separated at least by a given number of days off or assignments to a daytime shift.
- N10. Minimum Rest Time:** A minimum break specified in terms of minutes, must be guaranteed after a given number of consecutive working shifts. In detail, there must be a break of at least 11 h between two consecutive working shifts.
- N11. Period Succession:** There are forbidden period successions (e.g., *Late/ Early*) that cannot be assigned to the same nurse for two consecutive days.
- N12. Preferences:** A request of pre-assignment or shift-off with level *high*, *medium*, *low* is considered a preference that should be respected in the schedule.
- N13. Standard Shift And Skill:** Each nurse has a primary skill and a standard shift that should be preferred in the assignments.
- N14. Workteam:** Nurses belonging to the same workteam should work together in the same period, with different levels of priority (*high*, *medium*, *low*).
- N15. Minimum Days Off after Nights:** There must be a minimum of two days off after a sequence of night shifts.

*Patient oriented constraints.*

- P1. Optimal Staffing:** For each day, for each skill and for each period, the number of assigned nurses must be at least equal to the optimal coverage requirements.

This partition in three main categories is not a prescriptive one, since each healthcare institution can have a different view, but it offers the advantage to help schedulers to group constraints with the same stakeholder, highlight trade-offs and easily set their relative importance in order to obtained solutions that are more acceptable in practice given that they guarantee a fair balance between categories.

As customary, constraints are, in turn, split into hard ones that cannot be violated and soft ones that are penalized in the objective function. In our problem, the constraints that must be always satisfied are: A3, A4, A6, N3, N10, N11, N15. For the other ones, there is no distinction a priori between hard and soft constraints, but in each problem instance every single constraint can be specified as belonging to any of the two types (hard or soft).

The weights of the soft components are not fixed at the global level, but vary from instance to instance and are included in the input file of each instance (within the JSON attribute *Weights*).

According to the  $\alpha|\beta|\gamma$  classification introduced by De Causmaecker and Vanden Berghe [7], this problem can be classified as ASCN|RVNO|MG. Notice that, as remarked by Smet et al. [8] and Kingston [9], in practice it is crucial to use a consistent evaluation function of consecutive constraints that overlap between previous, current and upcoming planning horizon. As a consequence, our input data contains also some information about the previous roster (history data).

### 3. Mathematical model

In this section we present a MIP model for the problem. The input data and the notation employed in the model are summarized in Table 1.

The main decision variables are the binary four indexed  $x_{ndsk}$  that state whether nurse  $n$  on day  $d$  has been assigned to shift  $s$  with skill  $k$ . Additionally, other auxiliary variables have been defined and reported in Table 2.

We first describe the hard constraints of the problem and then deal with the other ones. The compatibility A3 of shift/skill, nurse/shift and nurse/skill are enforced with Eqs. (1), (2) and (3) respectively.

$$x_{ndsk} = 0 \quad \forall n \in N, d \in D, \langle s, k \rangle \in A \quad (1)$$

$$x_{ndsk} = 0 \quad \forall n \in N, d \in D, s \in S \setminus S^n, k \in K \quad (2)$$

$$x_{ndsk} = 0 \quad \forall n \in N, d \in D, s \in S, k \in K \setminus K^n \quad (3)$$

Constraints (4) ensure that a nurse is assigned at most to one shift per day (A4).

$$\sum_{s \in S} \sum_{k \in K} x_{ndsk} \leq 1 \quad \forall n \in N, d \in D \quad (4)$$

Constraints A6 about the minimum and maximum working time for a nurse in the planning horizon are guaranteed by inequalities (5)–(6).

$$\sum_{d \in D} \sum_{s \in S} \sum_{k \in K} \lambda_s \cdot x_{ndsk} + |\Phi_n^h| \cdot l_n \geq \Delta_n^{\min} \quad \forall n \in N \quad (5)$$

$$\sum_{d \in D} \sum_{s \in S} \sum_{k \in K} \lambda_s \cdot x_{ndsk} + |\Phi_n^h| \cdot l_n \leq \Delta_n^{\max} \quad \forall n \in N \quad (6)$$

Eqs. (7)–(9) satisfy the fixed requests of pre-assignment (7), period-off (8) or holiday (9) (constraint N3). The fixed request of working (any shift) in a day is imposed in Eqs. (10).

$$x_{ndsk} = 1 \quad \forall \langle n, d, s, k \rangle \in \Phi^a \quad (7)$$

$$\sum_{k \in K} \sum_{s \in S} x_{ndsk} = 0 \quad \forall \langle n, d, p \rangle \in \Phi^o \quad (8)$$

$$\sum_{s \in S} \sum_{k \in K} x_{ndsk} = 0 \quad \forall \langle n, d \rangle \in \Phi_n^h \quad (9)$$

$$\sum_{s \in S} \sum_{k \in K} x_{ndsk} = 1 \quad \forall \langle n, d \rangle \in \Phi^w \quad (10)$$

The linking constraints between the  $x_{ndsk}$  variables and the variables  $t_{nd}$  are reported in Eqs. (11), where the constant 1440 represents the number of minutes in a day ( $24 \times 60$ ). The  $t_{nd}$  variable indicates the start time of work for nurse  $n$  on day  $d$ , 0 in case of day-off.

$$t_{nd} = \sum_{s \in S} \sum_{k \in K} x_{ndsk} (s_s + 1440d - 1440) \quad \forall n \in N, d \in D \quad (11)$$

**Table 1**  
Notation for the input data.

| Symbol                           | Definition                                                                                                                            |
|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| $N$                              | set of nurses                                                                                                                         |
| $D$                              | set of days, with $D^+$ being the set of days including the last day of the previous planning horizon                                 |
| $D^n$                            | set of working days for nurse $n$                                                                                                     |
| $W$                              | set of Saturdays                                                                                                                      |
| $\mathcal{H}$                    | set of public holidays                                                                                                                |
| $P$                              | set of periods                                                                                                                        |
| $S$                              | set of shifts, with $\tilde{S} \subset S$ being the set of night shifts and $\hat{S} \subset S$ the set of day shifts                 |
| $S^n$                            | set of shifts of nurse $n$                                                                                                            |
| $S^p$                            | set of shifts of period $p$ ( $\cup_{p \in P} S^p = S$ )                                                                              |
| $K$                              | set of skills                                                                                                                         |
| $K^n$                            | set of skills of nurse $n$                                                                                                            |
| $A$                              | set of pairs $(s, k)$ of incompatible shift/skill                                                                                     |
| $C_{dpk}^{\min/\text{opt}/\max}$ | minimum/optimal/maximum number of nurse required on day $d$ for period $p$ and skill $k$                                              |
| $F$                              | set of pairs $\langle p_1, p_2 \rangle$ of forbidden period successions $p_1 \rightarrow p_2$                                         |
| $\lambda_s$                      | duration of shift $s$                                                                                                                 |
| $s_s$                            | start time of shift $s$                                                                                                               |
| $\Lambda_n^{\min/\max}$          | minimum/maximum working time for nurse $n$                                                                                            |
| $l_n$                            | average working time of nurse $n$ established in the contract                                                                         |
| $T_n^{\min/\max}$                | minimum/maximum number of working shifts for nurse $n$                                                                                |
| $W_n^{\max}$                     | maximum number of working weekends for nurse $n$                                                                                      |
| $\gamma_n^{\max}$                | maximum number of working Sundays for nurse $n$                                                                                       |
| $H_n^{\max}$                     | maximum number of working public holidays for nurse $n$                                                                               |
| $\gamma_n$                       | 1, if the complete weekend constraint is active for nurse $n$ , 0 otherwise                                                           |
| $c^{\min}$                       | minimum number of days off after a series of consecutive assignments                                                                  |
| $\Pi^a$                          | set of tuples $\langle n, d, s, k, l \rangle$ of preferred shift $s$ with skill $k$ for nurse $n$ on day $d$ with priority weight $l$ |
| $\Pi^o$                          | set of tuples $\langle n, d, s, l \rangle$ of undesired shift $s$ for nurse $n$ on day $d$ with priority weight $l$                   |
| $\Pi^h$                          | set of tuples $\langle n, d, l \rangle$ of desired day-off for nurse $n$ on day $d$ with priority weight $l$                          |
| $\Pi^w$                          | set of tuples $\langle n, d, l \rangle$ of desired working day for nurse $n$ on day $d$ with priority weight $l$                      |
| $\Phi^a$                         | set of tuples $\langle n, d, s, k \rangle$ of fixed requests of pre-assignments of shift $s$ with skill $k$ for nurse $n$ on day $d$  |
| $\Phi^o$                         | set of tuples $\langle n, d, p \rangle$ of fixed requests of not working any shift of period $p$ on day $d$ for nurse $n$             |
| $\Phi^w$                         | set of pairs $\langle n, d \rangle$ of fixed requests of working (any shift) for nurse $n$ on day $d$                                 |
| $\Phi_n^h$                       | set of fixed requests of vacation days of nurse $n$                                                                                   |
| $I$                              | set of tuples $\langle n_1, n_2, l \rangle$ of incompatible nurses $(n_1, n_2)$ with priority weight $l$                              |
| $E$                              | set of tuples $\langle n_1, n_2, l \rangle$ of preferred pairs of nurses $(n_1, n_2)$ with priority weight $l$                        |
| $\sigma_n^s$                     | standard shift for nurse $n$                                                                                                          |
| $\sigma_n^k$                     | standard skill for nurse $n$                                                                                                          |
| $\omega^i$                       | weight for violating constraint $i$                                                                                                   |
| $\omega_s^{A1}$                  | weight for violating constraint A1 for shift $s$                                                                                      |
| $WT$                             | maximum possible work team score                                                                                                      |
| $M$                              | large number                                                                                                                          |

Constraints (12) ensure that at least a rest time of 660 min (11 h) is guaranteed between two consecutive working shifts (N10).

$$t_{n,d+1} - \left( t_{nd} + \sum_{s \in S} \sum_{k \in K} \lambda_s x_{ndsk} \right) \geq 660 \sum_{s \in S} \sum_{k \in K} x_{ndsk} - M \left( 1 - \sum_{s \in S} \sum_{k \in K} x_{n,d+1,sk} \right) \quad \forall n \in N, d \in D^+ \setminus \{|D|\} \quad (12)$$

**Table 2**  
Notation for the decision variables.

| Symbol                                    | Definition                                                                                                                  |
|-------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| $x_{ndsk} \in \{0, 1\}$                   | 1 if nurse $n$ works shift $s$ with skill $k$ on day $d$ , 0 otherwise                                                      |
| $q_{nw} \in \{0, 1\}$                     | 1 if nurse $n$ works any shift on weekend $w$ , 0 otherwise                                                                 |
| $t_{nd} \geq 0$                           | start working time on day $d$ of nurse $n$ if she/he works any shift, 0 otherwise                                           |
| $y_{dpk}^i \in \mathbb{Z}^+$              | violations of the soft constraint $i \in \{A1, A2, P1\}$ for day $d$ , period $p$ , skill $k$                               |
| $y_{n1}^{A5 \min/\max} \in \mathbb{Z}^+$  | violations of the soft constraint A5 for nurse $n$                                                                          |
| $y_{nw}^{N1} \in \mathbb{Z}^+$            | violations of the soft constraint N1 for nurse $n$ on weekend $w$                                                           |
| $y_{nds}^{N2 \min/\max} \in \mathbb{Z}^+$ | violations of the soft constraint N2 of consecutive assignments of shift $s$ for nurse $n$ starting on day $d$              |
| $y_{nd}^{N2 \min/\max} \in \mathbb{Z}^+$  | violations of the soft constraint N2 of consecutive assignments (any shift) for nurse $n$ starting on day $d$               |
| $y_{nd}^{N2\_min/\max} \in \mathbb{Z}^+$  | violations of the soft constraint N2 of consecutive days off for nurse $n$ starting on day $d$                              |
| $y_{n_1 n_2 dp} \in \{0, 1\}$             | violations of the soft constraint $i \in \{N4, N14\}$ about nurses $n_1$ and $n_2$ on day $d$ , period $p$                  |
| $y_n^i \in \mathbb{Z}^+$                  | violations of the soft constraint $i \in \{N5, N6\}$ for nurse $n$                                                          |
| $y_{nd}^{N8} \in \mathbb{Z}^+$            | violations of the soft constraint N8 of minimum days off after consecutive assignments (any shift) for nurse $n$ on day $d$ |
| $y_{nd}^{N9} \in \mathbb{Z}^+$            | violations of the soft constraint N9 of minimum distance between two night blocks for nurse $n$ on day $d$                  |
| $y_{ndsk}^{N12^a} \in \mathbb{Z}^+$       | violations of the soft constraint N12 of preferred shift $s$ with skill $k$ for nurse $n$ on day $d$                        |
| $y_{nds}^{N12^o} \in \mathbb{Z}^+$        | violations of the soft constraint N12 of undesired shift $s$ for nurse $n$ on day $d$                                       |
| $y_{nd}^{N12^h} \in \mathbb{Z}^+$         | violations of the soft constraint N12 of desired day-off for nurse $n$ on day $d$                                           |
| $y_{nd}^{N12^w} \in \mathbb{Z}^+$         | violations of the soft constraint N12 of desired working day for nurse $n$ on day $d$                                       |
| $y_n^{N13^i} \in \mathbb{Z}^+$            | violations of the soft constraint N13 for nurse $n$ of standard shift/skill $i \in \{s, k\}$                                |

The compliance with the period succession constraint N11 is obtained through the constraints (13), where  $F$  represents the set of pairs  $\langle p_1, p_2 \rangle$  such that a shift of period  $p_1$  must not be followed by a shift of period  $p_2$ .

$$\sum_{k \in K} \sum_{s_1 \in S^{p_1}} x_{nds_1 k} + \sum_{k \in K} \sum_{s_2 \in S^{p_2}} x_{n,d+1,s_2 k} \leq 1 \quad \forall n \in N, d \in D^+ \setminus \{|D|\}, \langle p_1, p_2 \rangle \in F \quad (13)$$

Finally, constraints (14) impose a minimum of two resting days after a single night shift or a block of consecutive night shifts (N15).

$$\sum_{k \in K} \sum_{s \in \tilde{S}} 2x_{ndsk} + \sum_{k \in K} \sum_{s \in \tilde{S}} (x_{n,d+1,sk} + x_{n,d+2,sk}) \leq 2 \quad n \in N, d \in D^+ \setminus \{|D|, |D| - 1\} \quad (14)$$

The remaining constraints are soft ones, so that they can be violated but a corresponding slack/surplus variable  $y$  is activated and then penalized in the objective function (34).

The inequalities (15)–(17) model the range constraints about the minimum/optimal/maximum number of nurse per day/period/skill: understaffing A1 (15), optimal staffing P1 (16) and exceeding staffing A2 (17).

$$\sum_{n \in N} \sum_{s \in S^p} x_{ndsk} + y_{dpk}^{A1} \geq C_{dpk}^{\min} \quad \forall d \in D, p \in P, k \in K \quad (15)$$

$$\sum_{n \in N} \sum_{s \in S^p} x_{ndsk} + y_{dpk}^{P1} \geq C_{dpk}^{\text{opt}} \quad \forall d \in D, p \in P, k \in K \quad (16)$$

$$\sum_{n \in N} \sum_{s \in SP} x_{ndsk} - y_{dpk}^{A2} \leq C_{dpk}^{\max} \quad \forall d \in D, p \in P, k \in K \quad (17)$$

Ranged constraints (18) and (19) model the minimum and maximum number of assignments in the horizon (A5).

$$\sum_{d \in D} \sum_{s \in S} \sum_{k \in K} x_{ndsk} + \sum_{d \in \Phi_n^h \cap D^n} (1 - \sum_{s \in S} \sum_{k \in K} x_{ndsk}) + y_n^{A5 \min} \geq T_n^{\min} \quad \forall n \in N \quad (18)$$

$$\sum_{d \in D} \sum_{s \in S} \sum_{k \in K} x_{ndsk} + \sum_{d \in \Phi_n^h \cap D^n} (1 - \sum_{s \in S} \sum_{k \in K} x_{ndsk}) - y_n^{A5 \max} \leq T_n^{\max} \quad \forall n \in N \quad (19)$$

For the complete weekend constraint N1, first the channeling constraints between  $x_{ndsk}$  and  $q_{nw}$  variables are set; the variable  $q_{nw}$  is activated if nurse  $n$  works at least one day on the weekend starting on day  $w$  (20). Then the N1 constraints are imposed through inequalities (21), where the complete weekend constraint is active for nurse  $n$  if the parameter  $\gamma_n$  is equal to 1.

$$\sum_{s \in S} \sum_{k \in K} (x_{nws} + x_{n,w+1,s}) \leq 2q_{nw} \quad \forall n \in N, w \in W \quad (20)$$

$$\sum_{s \in S} \sum_{k \in K} (x_{nws} + x_{n,w+1,s}) + y_{nw}^{N1} \geq 2\gamma_n \cdot q_{nw} \quad \forall n \in N, w \in W \quad (21)$$

The incompatibility N4 and workteam N14 constraints are formulated similarly in inequalities (22) and (23)–(24) considering the tuples  $(n_1, n_2, l)$  of incompatible/preferable nurses with a given level of priority  $l$ . Notice that in the objective function (34) the cost component related to N14 is subtracted to the constant WT which is the maximum possible workteam score.

$$\sum_{s \in SP} \sum_{k \in K} (x_{n_1dsk} + x_{n_2dsk}) - y_{n_1n_2dp}^{N4} \leq 1 \quad \forall \langle n_1, n_2 \rangle \in I, \forall d \in D, p \in P \quad (22)$$

$$\sum_{s \in SP} \sum_{k \in K} (x_{n_1dsk} + x_{n_2dsk}) - y_{n_1n_2dp}^{N14} \leq 1 \quad \forall \langle n_1, n_2 \rangle \in E, \forall d \in D, p \in P \quad (23)$$

$$y_{n_1n_2dp}^{N14} \leq \sum_{s \in SP} \sum_{k \in K} x_{n_1dsk} \quad \wedge \quad y_{n_1n_2dp}^{N14} \leq \sum_{s \in SP} \sum_{k \in K} x_{n_2dsk} \quad \forall \langle n_1, n_2 \rangle \in E, \forall d \in D, p \in P \quad (24)$$

Constraints (25) limit the maximum number of working weekends (N5), constraints (26) limit the maximum number of working Sundays (N6), and constraints (27) limit the maximum number of working public holidays (N7).

$$\sum_{w \in W} q_{nw} - y_n^{N5} \leq W_n^{\max} \quad \forall n \in N \quad (25)$$

$$\sum_{w \in W} \sum_{s \in S} \sum_{k \in K} x_{n,w+1,sk} - y_n^{N6} \leq \gamma_n^{\max} \quad \forall n \in N \quad (26)$$

$$\sum_{h \in \mathcal{H}} \sum_{s \in S} \sum_{k \in K} x_{nhsk} - y_n^{N7} \leq H_n^{\max} \quad \forall n \in N \quad (27)$$

Preferred shifts (N12) are modeled in constraints (28), while shift-off, day-off and working day requests of level different than fixed are imposed in (29), (30) and (31), respectively.

$$(1 - x_{ndsk}) = y_{ndsk}^{N12^a} \quad \forall \langle n, d, s, k \rangle \in \Pi^a \quad (28)$$

$$\sum_{k \in K} x_{ndsk} = y_{nds}^{N12^o} \quad \forall \langle n, d, s \rangle \in \Pi^o \quad (29)$$

$$\sum_{k \in K} \sum_{s \in S} x_{ndsk} = y_{nd}^{N12^h} \quad \forall \langle n, d \rangle \in \Pi^h \quad (30)$$

$$\sum_{k \in K} \sum_{s \in S} x_{ndsk} + y_{nd}^{N12^w} = 1 \quad \forall \langle n, d \rangle \in \Pi^w \quad (31)$$

Any assignment different than the standard shift  $\sigma_n^s$  or standard skill  $\sigma_n^k$  (N13) is taken into account in Eqs. (32) and (33).

$$\sum_{d \in D} \sum_{s \in S \setminus \{\sigma_n^s\}} \sum_{k \in K} x_{ndsk} = y_n^{N13^s} \quad \forall n \in N \quad (32)$$

$$\sum_{d \in D} \sum_{s \in S} \sum_{k \in K \setminus \{\sigma_n^k\}} x_{ndsk} = y_n^{N13^k} \quad \forall n \in N \quad (33)$$

Constraints concerning consecutive assignments/days off (N2, N8, N9) are formulated as in the model presented by Mischek and Musliu [10, Eqs.(7)–(21)] and are not reported here for brevity. Their violation is taken into account through corresponding variables  $y^{N2}, y^{N8}, y^{N9}$  employed in the objective function.

The objective function (34) is the weighted sum of the violations of soft constraints A1–7, N1–14 and P1.

$$\begin{aligned} \min z = & \omega^{A1} \sum_{d \in D, p \in P, k \in K} \omega_s^{A1} \cdot y_{dpk}^{A1} + \sum_{i \in \{A2, P1\}} \omega^i \sum_{d \in D, s \in S, k \in K} y_{dsk}^i + \\ & \sum_{i \in \{A5^{\min}, A5^{\max}, N5, N6, N7, N13^s, N13^k\}} \omega^i \sum_{n \in N} y_n^i + \omega^{N1} \sum_{n \in N, w \in W} y_{nw}^{N1} \\ & + \sum_{i \in \{N2^{\min}, N2^{\max}\}} \omega^{N2} \sum_{n \in N, d \in D, s \in S} y_{nds}^i \\ & + \sum_{i \in \{N2^{\min}, N2^{\max}, N2^{\min}, N2^{\max}\}} \omega^{N2} \sum_{n \in N, d \in D} y_{nd}^i + \\ & \omega^{N4} \sum_{\langle n_1, n_2, l \rangle \in I} \sum_{d \in D, p \in P} l \cdot y_{n_1n_2dp}^{N4} + \sum_{i \in \{N8, N9\}} \omega^i \sum_{n \in N, d \in D} y_{nd}^i + \\ & \omega^{N12} \left( \sum_{\langle n, d, s, k, l \rangle \in \Pi^a} l \cdot y_{ndsk}^{N12^a} + \sum_{\langle n, d, s, l \rangle \in \Pi^o} l \cdot y_{nds}^{N12^o} \right. \\ & \left. + \sum_{\langle n, d, l \rangle \in \Pi^h} l \cdot y_{nd}^{N12^h} + \sum_{\langle n, d, l \rangle \in \Pi^w} l \cdot y_{nd}^{N12^w} \right) + \\ & \omega^{N14} \left( WT - \sum_{\langle n_1, n_2, l \rangle \in E} \sum_{d \in D, p \in P} l \cdot y_{n_1n_2dp}^{N14} \right) \end{aligned} \quad (34)$$

#### 4. Related work

Nurse rostering is one of the most studied optimization problems in the scientific literature. First works appeared in the 70s and since then, many papers have been dedicated to this topic due to both the complexity of the problem, that makes it still challenging for researchers, and its social and economical impact for healthcare organizations that need optimization solutions that permits to rationalize costs and guaranteed patient satisfaction and personnel well-being. Despite the extensive literature on this problem (for a comprehensive review see the surveys [1,2]), there is still a gap between the problems proposed in the literature and those arising in real-world, due to the fact that each healthcare institution (or even ward in the same organization) has its own features, rules and practices. Indeed, as reported by Van den Bergh et al. [1] less than 15% of solution approaches proposed in the literature were implemented in practice in healthcare systems, so that commercial solvers have still not benefited from the advances obtained by the research community in developing effective methods [5,11].

In this review, we consider approaches to model and solve real-world NR problems. Since the focus is on complete formulations, we have excluded any paper that addresses simplified/academic version of the problem.

As demonstrated by Smet [12], constraints concerning consecutiveness make NR problems particularly difficult to solve, so that large instances are not within the reach of standard time-indexed IP models. In order to make exact methods competitive also for real-world NR problems, more sophisticated modeling techniques, such as decompositions [13] and reformulations [10, 12, 14, 15], are necessary. Despite such effort, for large instances, the best results in the literature have been usually obtained using non exact methods, like for example the hybrid method proposed by Legrain et al. [16] for the static version of the INRC-II problem that integrates a *branch and price* procedure in an Adaptive Large Neighborhood Search framework. As a consequence, heuristic techniques, such as metaheuristics [17–22], hyper-heuristics [8], or hybrid approaches [23–25] are usually implemented.

In particular, Knust and Xie [22] recently proposed a Simulated Annealing algorithm for the solution of a NR problem, which is a simplified version of our problem. For example, in our model shifts depend on the skill, and each nurse has a subset of feasible shifts whom can be assigned to. In addition, more complex consecutiveness constraints (N8, N9, N10, N15) imposed by Italian regulations are considered. Regarding the SA algorithm, authors implemented neighborhood structures that change or swap nurses, similarly to the one described in Section 5.2. Their solver was tested on benchmark instances from the literature and on three cases from hospitals in Germany and Austria.

Finally, recent works by Böðvarsdóttir et al. [26] adopted a different research perspective by investigating how to deal with the multi-objective nature of NR problems. The authors present a general methodology to assist practitioners to set the weights of the different objective in case of single cost function expressed as the weighted sum of the multiple constraints violations. Conversely, Böðvarsdóttir et al. [27] introduced a multi-objective approach that guides the search towards acceptable compromise solutions, by automatically estimating acceptance thresholds.

## 5. Solution method

Our solution method is based on local search, we thus describe its key components: the search space, the cost function and the initial solution (Section 5.1), then the neighborhood relations (Section 5.2) and finally the metaheuristic algorithm, namely Simulated Annealing (Section 5.3).

### 5.1. Search space, cost function and initial solution

The search state is a direct representation of the nurses' roster, consisting of a sequence of shifts (or absences) assigned to each nurse on each day of the planning horizon. The state also includes the skill performed by a nurse on each working shift.

The cost function  $f$  is the weighted sum of all hard constraint violations, multiplied by a suitable hard weight, and soft penalties. As already mentioned in Section 2, the distinction between hard and soft constraints, as well as all weights of the objectives, are defined in the input data so that they can vary from one instance to another.

For the initial solution strategy, the procedure first populates the roster with fixed pre-assignments and shift-off requests of nurses. Then, an iterative procedure assigns, for each day and each period, nurses to shifts so as to satisfy the optimal coverage requirements. The nurse is selected randomly among those that have the necessary skill. To build the initial solution only standard shifts of a nurse are used for the assignments.

### 5.2. Neighborhood relations

The typical basic neighborhood relation in nurse rostering is the swapping of the shifts of two nurses on one day. We call this neighborhood SwapNurses:

**SwapNurses (SN)** Given a day  $d$ , a nurse  $n_1$  and a nurse  $n_2$ , such that  $n_1$  (resp.  $n_2$ ) is assigned to shift  $sh_1$  (resp.  $sh_2$ ) with skill  $sk_1$  (resp.  $sk_2$ ) on day  $d$ , the move  $SN(n_1, n_2, d)$  swaps the shifts of the two nurses on day  $d$ , i.e. assigns shift  $sh_2$  and skill  $sk_2$  to  $n_1$  and  $sh_1$  with  $sk_1$  to  $n_2$ .

*Preconditions:*

- i shift  $sh_2$  is feasible for nurse  $n_1$ ; shift  $sh_1$  is feasible for nurse  $n_2$ ;
- ii nurse  $n_1$  is endowed with skill  $sk_2$ ; nurse  $n_2$  is endowed with skill  $sk_1$ ;
- iii nurse  $n_1$  has not a fixed pre-assignment for shift  $sh_1$  (and skill  $sk_1$ ) on day  $d$ ; nurse  $n_2$  has not a fixed pre-assignment for shift  $sh_2$  (and skill  $sk_2$ ) on day  $d$ ;
- iv nurse  $n_1$  has not a fixed shift-off request for shift  $sh_2$  (and skill  $sk_2$ ) on day  $d$ ; nurse  $n_2$  has not a fixed shift-off request for shift  $sh_1$  (and skill  $sk_1$ ) on day  $d$ .

For our problem the personnel coverage is expressed in terms of minimum, optimal and maximum values, so that it is necessary to adopt also a neighborhood that changes the number of total working shifts. We thus introduce the ChangeNurse relation that modifies the assignment of a single nurse in a day:

**ChangeNurse (CN)** Given a day  $d$ , a nurse  $n$ , a shift  $sh$  and a skill  $sk$ , the move  $CN(n, d, sh, sk)$  assigns shift  $sh$  and skill  $sk$  to nurse  $n$  on day  $d$ . If  $sh$  is not a working shift, the skill  $sk$  is not significative.

*Preconditions:*

- i shift  $sh$  is feasible for nurse  $n$ ;
- ii nurse  $n$  is endowed with skill  $sk$  (if significative);
- iii nurse  $n$  has not a fixed pre-assignment for the current shift (and skill) on day  $d$ ;
- iv nurse  $n$  has not a fixed shift-off request for shift  $sh$  (and skill  $sk$ ) on day  $d$ .

We thus employ a new neighborhood which is the set-union of SwapNurses and ChangeNurse. At each iteration of the search method, a SN move is selected with probability  $p_{SN}$  and CN with probability  $1 - p_{SN}$ , where  $p_{SN}$  is a parameter of the method that has been properly tune (see Section 6.2).

For the CN neighborhood, the selection of the shift for a nurse is not uniform, but biased towards the assignments to standard shifts, in order to better fulfill constraint N13. This is obtained by adding the parameter  $k$  that represents the probability of selecting the standard shift. With probability  $1 - k$  we select any suitable shift with uniform distribution.

### 5.3. Simulated annealing

As a metaheuristic to guide the local search we use Simulated Annealing (SA) [28]. The SA procedure starts from an initial random state and draws, at each iteration, a random move in the set-union neighborhood as explained above. Calling  $\Delta f$  the difference of cost induced by the selected move, this is always accepted if  $\Delta f \leq 0$ , whereas it is accepted based on time-decreasing exponential distribution when  $\Delta f > 0$ . More precisely, a worsening move is accepted with probability  $e^{-\Delta f/T}$ , where  $T$  is the temperature. The temperature starts at the initial high value  $T_0$ ,

**Table 3**  
Features of the real-life data set.

| Ward | # Inst | Nurses | Shifts | Skills | Periods | Contracts |
|------|--------|--------|--------|--------|---------|-----------|
| W1   | 4      | 48     | 9      | 4      | 4       | 2         |
| W2   | 4      | 15     | 9      | 4      | 4       | 1         |
| W3   | 4      | 28     | 22     | 7      | 5       | 6         |
| W4   | 3      | 23–24  | 20     | 5      | 4       | 4         |
| W5   | 4      | 24     | 19     | 6      | 4       | 4         |
| W6   | 2      | 30–31  | 23     | 6      | 4       | 4         |
| W7   | 1      | 31     | 17     | 1      | 3       | 4         |
| W8   | 4      | 30–31  | 9      | 2      | 4       | 1         |
| W9   | 4      | 30–31  | 10     | 3      | 4       | 1         |
| W10  | 4      | 30–31  | 14     | 4      | 4       | 4         |

and decreases by being multiplied by  $\alpha$  (with  $0 < \alpha < 1$ ), each time a fixed number of samples  $N_s$  has been drawn.

To the basic SA procedure, we add the *cut-off* mechanism that speeds up the early stages [29]. The idea is to decrease the temperature as soon as a fixed number of moves has been accepted. That is, we add a new parameter  $N_a$ , that represents the maximum number of accepted moves at each temperature. The temperature is decreased when the first of the following two conditions occurs: (i) the number of accepted moves reaches  $N_a$ , (ii) the number of sampled moves reaches  $N_s$ .

We use the total number of iterations  $\mathcal{I}$  as stop criterion. This guarantees that the running time is the same for all configurations of the SA parameters. In order to keep total number of iterations  $\mathcal{I}$  fixed, one of the parameters, namely  $N_s$  is computed from  $\mathcal{I}$  and from the other parameters using the formula:

$$N_s = \mathcal{I} / \left( \frac{\log(T_f/T_0)}{\log \alpha} \right)$$

where  $T_f$  is the *expected* final temperature. Notice that  $T_f$  is used in the formula to compute  $N_s$ , but the actual final temperature might fall below, as the iterations saved in the early stages, due to the cut-off mechanism, are executed at the end of the search.

Given that  $N_s$  is not fixed,  $N_a$  is not fixed either, but it is set to be a fraction  $\rho$  of the computed  $N_s$ , where  $\rho$  is a new parameter and  $N_a = \rho N_s$ .

## 6. Experimental analysis

All experiments ran on an Ubuntu Linux 20.04.2 LTS machine with 64 GB of RAM and 32 AMD Ryzen Threadripper PRO 3975WX processors (3.50 GHz), hyper-threaded to 64 virtual cores. The MIP model has been implemented in IBM ILOG CPLEX (v. 12.10) using the Concert Technology, which provides the C++ interface.

A single virtual core has been dedicated to each local search experiment so that 64 experiments are run in parallel. Conversely, we ran one CPLEX execution at the time, and no limits have been imposed to the use of multiple cores or memory for CPLEX.

### 6.1. Dataset

At the moment, we have collected 34 instances coming from 10 different wards (belonging to different health facilities located in the North Italy), which exhibits variety of dimension, contracts, constraints, practices, and shift types.

Table 3 summarizes, for each ward, the values of the main features of the corresponding instances. The planning horizon is equal of one month for all instances. Instances, conveniently anonymized, are available at <https://bitbucket.org/satt/nrp-instances> for future comparisons.

**Table 4**  
Parameter settings.

| Name     | Description                      | Range     | Value  |
|----------|----------------------------------|-----------|--------|
| $T_0$    | Start temperature                | 500–1500  | 907.41 |
| $T_f$    | Minimum temperature              | 1–2       | 1.92   |
| $\alpha$ | Cooling rate                     | 0.8–0.999 | 0.955  |
| $\rho$   | Cutoff ratio                     | 0.1–0.2   | 0.16   |
| $p_{SN}$ | Probability of a SwapNurses      | 0.4–0.7   | 0.49   |
| $k$      | Probability of a std shift/skill | 0.6–0.9   | 0.83   |

### 6.2. Parameter tuning

The parameters of the solution methods are those of the SA algorithm ( $T_0$ ,  $T_f$ ,  $\alpha$ ,  $\rho$ ) and those related to the neighborhoods ( $p_{SN}$  and  $k$ ). Their tuning has been performed using the tool json2run [30] which samples the configurations using the Hammersley point set [31] and implements the F-Race procedure [32] for comparing them with  $p$ -value = 0.05.

The parameter values of the best configuration for our SA are shown in Table 4, which also reports the initial ranges defined with preliminary experiments. For the tuning phase the total number of iterations was set to  $\mathcal{I} = 10^6$ .

### 6.3. Experimental results

This section presents the computational results on the real-world instances whose main features are described in Section 6.1.

In order to reproduce the practical usage of the application, we experimented two different timeouts: short runs with a total number of iterations  $\mathcal{I}$  equal to  $10^6$  corresponding to an average computational time of 240 s, and long runs with  $\mathcal{I} = 10^7$  (corresponding to about 2400 s). Indeed, although the roster is usually scheduled on a monthly basis, it is essential for practitioners to obtain quickly (e.g., in a few minutes) a good solution, which is immediately evaluated to identify missing data, inconsistencies, errors. The system can also be used in a batch mode, such that solver is launched with a longer timeout. However, in our experience, also in this case the running time must not exceed a couple of hours in order to be considered reasonable by schedulers.

For comparison, we also implemented a simple *Steepest Descent* (SD) technique that at each iteration explores the set-union neighborhood exhaustively, selects the best neighbor, and stops as soon as it reaches a local minimum.

Table 5 shows the minimum cost, the average cost, the deviation, and average running time of 30 repetitions of SA for each problem instance. The column SD reports the value obtained by the deterministic SD procedure, with an average running time of 868.2s. The last column shows the lower bounds obtained by running the MIP model with a timeout of 7200s. In order to have CPLEX running within reasonable computational time we had to remove all constraints about consecutiveness from the model, namely N2, N8, and N9.

A solution obtained for this relaxed problem thus provides a lower bound and not an optimal value, even when the solver stops before the timeout and delivers the optimal solution for the relaxed problem. The optimality for the complete problem is guaranteed only for the few instances for which such lower bound is equal to the corresponding upper bound found by SA. The optimal values are underlined, whereas the symbol – denotes that CPLEX ran out of memory.

We can see that the deviation of SA is rather low, demonstrating that the solution method is quite robust, in particular for the longer timeout. The average improvement obtained by longer runs of SA compared to short ones varies quite substantially among different wards (from 0.01% of W5-01 to 24.74% of W10-02), suggesting that some cases are strongly bounded, while for

**Table 5**  
Comparative results.

| Instance | SA short runs |         |       |       | SA long runs |         |      |        | SD    | LB      |
|----------|---------------|---------|-------|-------|--------------|---------|------|--------|-------|---------|
|          | z             |         |       | t [s] | z            |         |      | t [s]  | z     | z       |
|          | min           | avg     | dev   | avg   | min          | avg     | dev  | avg    |       |         |
| W1-01    | 1865          | 2059.5  | 90.7  | 492.6 | 1691         | 1744.3  | 51.8 | 4922.2 | 13414 | 945.8   |
| W1-02    | 1882          | 2061.3  | 91.1  | 525.1 | 1593         | 1663.7  | 38.0 | 5244.3 | 3263  | 833.0   |
| W1-03    | 2505          | 2636.9  | 84.6  | 425.0 | 2297         | 2364.8  | 35.3 | 4238.6 | 4208  | 1758.8  |
| W1-04    | 2192          | 2328.4  | 85.7  | 444.1 | 1998         | 2027.6  | 22.2 | 4432.3 | 4129  | 1293.1  |
| W2-01    | 2448          | 2632.4  | 86.8  | 76.7  | 2258         | 2360.5  | 50.1 | 759.9  | 5036  | 1433.0  |
| W2-02    | 2646          | 2776.4  | 83.5  | 82.0  | 2406         | 2512.0  | 58.6 | 817.0  | 5440  | –       |
| W2-03    | 2904          | 3102.6  | 95.9  | 78.0  | 2725         | 2826.7  | 72.8 | 779.2  | 5033  | 2167.4  |
| W2-04    | 1992          | 2127.6  | 80.2  | 84.2  | 1738         | 1858.2  | 55.9 | 837.7  | 4073  | –       |
| W3-01    | 4216          | 4321.4  | 64.0  | 193.3 | 4116         | 4238.0  | 60.7 | 1930.4 | 5179  | 3830.0  |
| W3-02    | 2983          | 3095.3  | 57.4  | 195.9 | 2846         | 2984.9  | 61.8 | 1955.5 | 3688  | 2298.0  |
| W3-03    | 4929          | 5046.9  | 64.3  | 168.3 | 4885         | 4967.9  | 47.7 | 1682.4 | 6138  | 4499.0  |
| W3-04    | 4756          | 4879.5  | 79.7  | 185.1 | 4706         | 4792.6  | 58.7 | 1847.2 | 5529  | 4170.0  |
| W4-01    | 1866          | 1877.8  | 5.7   | 163.9 | 1860         | 1868.2  | 4.4  | 1633.6 | 2274  | 1805.4  |
| W4-02    | 2135          | 2146.9  | 6.3   | 156.8 | 2122         | 2133.7  | 4.0  | 1565.1 | 2517  | 2099.0  |
| W4-03    | 1964          | 1976.7  | 7.2   | 154.0 | 1955         | 1965.0  | 6.1  | 1538.6 | 2270  | 1915.0  |
| W5-01    | 18950         | 19078.0 | 63.8  | 138.7 | 18940        | 19076.0 | 72.7 | 780.7  | 19230 | 14185.0 |
| W5-02    | 6106          | 6135.6  | 35.2  | 143.6 | 6106         | 6111.2  | 2.1  | 755.8  | 6810  | 6106.0  |
| W5-03    | 25130         | 25207.3 | 76.3  | 127.6 | 25130        | 25135.7 | 17.4 | 1277.4 | 25475 | 23480.0 |
| W5-04    | 4081          | 4084.8  | 2.1   | 135.5 | 4077         | 4081.1  | 2.0  | 1359.2 | 4416  | 4077.0  |
| W6-01    | 5212          | 5229.0  | 19.5  | 241.0 | 5202         | 5215.1  | 18.5 | 2409.4 | 5242  | 5202.0  |
| W6-02    | 5532          | 5537.3  | 2.6   | 242.2 | 5517         | 5529.8  | 4.7  | 2423.4 | 5569  | 5517.0  |
| W7-01    | 80            | 88.2    | 4.4   | 88.8  | 65           | 75.7    | 4.1  | 887.7  | 110   | 50.0    |
| W8-01    | 6830          | 6919.0  | 40.0  | 146.5 | 6820         | 6848.3  | 30.9 | 1467.1 | 8020  | 6620.0  |
| W8-02    | 1550          | 1578.8  | 23.0  | 157.6 | 1547         | 1553.6  | 6.3  | 1576.8 | 2382  | 1267.0  |
| W8-03    | 1553          | 1612.7  | 31.5  | 149.1 | 1535         | 1552.9  | 14.3 | 1489.3 | 2749  | 1311.8  |
| W8-04    | 1577          | 1606.7  | 26.3  | 162.1 | 1574         | 1576.6  | 2.4  | 1621.1 | 2478  | 1114.0  |
| W9-01    | 3353          | 3507.8  | 102.8 | 256.8 | 2917         | 3037.3  | 61.3 | 2548.8 | 5963  | 2313.9  |
| W9-02    | 3802          | 3971.4  | 93.8  | 282.8 | 3364         | 3480.9  | 64.9 | 2805.1 | 6339  | 2172.5  |
| W9-03    | 5247          | 5491.8  | 123.8 | 232.8 | 4718         | 4844.3  | 67.3 | 2306.5 | 8181  | 3443.8  |
| W9-04    | 4345          | 4561.7  | 112.8 | 241.6 | 3944         | 4038.2  | 69.3 | 2405.8 | 6777  | 2584.0  |
| W10-01   | 1825          | 2045.2  | 125.3 | 499.4 | 1521         | 1551.1  | 26.9 | 4990.0 | 4201  | 479.0   |
| W10-02   | 1894          | 2135.5  | 144.6 | 533.9 | 1581         | 1607.5  | 22.5 | 5329.7 | 4639  | 451.0   |
| W10-03   | 6828          | 7033.6  | 92.0  | 389.1 | 6640         | 6758.4  | 67.4 | 3895.1 | 19348 | 6064.1  |
| W10-04   | 2262          | 2435.6  | 116.1 | 473.5 | 1880         | 1981.3  | 46.6 | 4726.5 | 4454  | 1146.6  |

others there is room for improvements allowing more time for the search.

We also notice that the results obtained by SD are much worse than the ones for SA, demonstrating that the use of a smarter metaheuristic such as SA is necessary. Finally, we see that SA is able to find an optimal solution for a few instances, matching the lower bounds obtained by CPLEX. In other cases, it is rather far from the lower bound, although it is not known whether they are tight enough.

In Table 6, we report the average costs for the objective components introduced in Section 2. We show only the components that have a positive cost, the others are neglected as they are consistently equal to zero. We see that the cost is dominated by the components A1 (Understaffing), N2 (Consecutiveness), N13 (Standard Shift and Skill), N14 (Workteam), and P1 (Optimal staffing), which clearly prevail on all the others.

The components A1 and P1 are both related to the level of staffing, and their high cost reflects the current shortage of hospitals in terms of nurses. The high value of the component N2 is also not surprising as keeping the correct shift successions is normally a concern in hospitals. Finally, the high values of components N13 and N14 show that it is not possible to give the standard shift and the best team to each nurse all the time.

## 7. Conclusions and future work

We defined a complex nurse rostering problem that considers all constraints, coming from real-world situations. We present a MIP model and developed a metaheuristic solution method to

solve this rich problem, which has been able to find good quality solutions for the available instances in reasonable running times. We are now ready to enter the stage of validating the solutions with the hospital management.

The solver is a significant extension on our previous work [6] on a simplified problem, which obtained state-of-the-art results for benchmarks of the Second International Nurse Rostering Competition (INRC-II). All instances are available to be used for future researches and comparisons.

During this research project, we have identified several future opportunities to further improve and refine our work.

- Following Böðvarsdóttir et al. [26], it would be very useful to devise a more sophisticated method with respect to our division in stakeholders' categories to help practitioners to automatically set the weights of the different components of the cost function.
- Some critical care units are characterized by a high level of work-related stress (demanding workload, fear of being infected, dying patients, end-of-life decision, ...) that could lead on the one hand to higher risk of inhospital mortality of patients, on the other hand to burnout syndrome for staffing [33]. As a consequence, for this situations it would be necessary to include an explicit consideration of well-being measures in the objective function.
- As pointed out by Wickert et al. [34], the real working situation of nurses is particularly dynamic due to absenteeism and demand peaks, so that planned roster are continuously affected by changes. In this context, it would be important

**Table 6**  
Average values of the different objectives for long runs.

| Instance | A1     | A2    | A5    | A6   | N2     | N4  | N8  | N12  | N13   | N14    | P1      |
|----------|--------|-------|-------|------|--------|-----|-----|------|-------|--------|---------|
| W1-01    | 0.0    | 0.0   | 25.4  | 0.0  | 310.0  | 0.3 | 0.0 | 7.3  | 266.3 | 464.9  | 422.0   |
| W1-02    | 0.0    | 0.0   | 40.4  | 0.0  | 385.0  | 0.3 | 0.0 | 13.0 | 296.7 | 341.7  | 383.8   |
| W1-03    | 3.3    | 0.0   | 16.5  | 0.0  | 320.0  | 0.1 | 0.0 | 12.0 | 316.7 | 777.7  | 795.8   |
| W1-04    | 0.0    | 0.0   | 17.1  | 0.0  | 350.0  | 0.3 | 0.0 | 4.9  | 265.0 | 791.6  | 510.7   |
| W2-01    | 120.0  | 0.0   | 13.9  | 0.0  | 606.7  | 0.0 | 0.0 | 0.0  | 50.7  | 0.0    | 1385.3  |
| W2-02    | 116.7  | 0.0   | 16.0  | 0.0  | 743.3  | 0.0 | 0.0 | 0.0  | 47.7  | 0.0    | 1420.3  |
| W2-03    | 814.7  | 0.0   | 10.4  | 0.0  | 438.3  | 0.0 | 0.0 | 0.0  | 89.0  | 0.0    | 1474.3  |
| W2-04    | 0.0    | 0.0   | 12.6  | 0.0  | 476.7  | 0.0 | 0.0 | 0.0  | 22.8  | 0.0    | 1183.5  |
| W3-01    | 2400.0 | 0.0   | 16.2  | 0.0  | 318.3  | 0.0 | 0.0 | 0.0  | 7.8   | 0.0    | 1413.0  |
| W3-02    | 2250.0 | 0.0   | 11.6  | 0.0  | 0.0    | 0.0 | 0.0 | 0.0  | 0.0   | 0.0    | 723.3   |
| W3-03    | 2516.7 | 0.0   | 14.6  | 0.0  | 356.7  | 0.0 | 0.0 | 0.0  | 5.8   | 0.0    | 2002.2  |
| W3-04    | 2446.7 | 0.0   | 15.9  | 0.0  | 485.0  | 0.0 | 0.0 | 0.0  | 0.7   | 0.0    | 1759.0  |
| W4-01    | 0.0    | 0.0   | 16.7  | 0.0  | 0.0    | 0.0 | 0.0 | 0.0  | 200.8 | 0.0    | 1650.7  |
| W4-02    | 0.0    | 0.0   | 26.0  | 0.0  | 0.0    | 0.0 | 0.0 | 0.0  | 213.5 | 0.0    | 1894.2  |
| W4-03    | 0.0    | 0.0   | 25.7  | 0.0  | 0.0    | 0.0 | 0.0 | 0.0  | 248.3 | 0.0    | 1691.0  |
| W5-01    | 861.3  | 0.0   | 112.0 | 0.0  | 3313.3 | 0.0 | 0.0 | 0.0  | 99.5  | 0.0    | 14704.0 |
| W5-02    | 1760.0 | 0.0   | 1.0   | 0.0  | 0.0    | 0.0 | 0.0 | 0.0  | 110.0 | 0.0    | 4240.5  |
| W5-03    | 1651.7 | 200.0 | 10.0  | 30.0 | 1724.0 | 0.0 | 0.0 | 0.0  | 120.0 | 0.0    | 21400.0 |
| W5-04    | 0.0    | 80.0  | 1.4   | 0.0  | 0.0    | 0.0 | 0.0 | 0.0  | 100.2 | 0.0    | 3899.5  |
| W6-01    | 0.0    | 0.0   | 32.1  | 0.0  | 6.7    | 0.0 | 0.0 | 0.0  | 835.0 | 0.0    | 4341.3  |
| W6-02    | 0.0    | 0.0   | 32.0  | 0.0  | 0.0    | 0.0 | 0.0 | 0.0  | 806.5 | 0.0    | 4691.3  |
| W7-01    | 0.0    | 0.0   | 0.0   | 0.0  | 0.0    | 0.0 | 0.0 | 0.0  | 0.0   | 0.0    | 75.7    |
| W8-01    | 0.0    | 0.0   | 20.0  | 0.0  | 49.7   | 0.0 | 0.0 | 0.0  | 0.0   | 0.0    | 6600.0  |
| W8-02    | 0.0    | 0.0   | 2.0   | 0.0  | 198.3  | 0.0 | 0.0 | 5.5  | 0.0   | 0.0    | 1265.2  |
| W8-03    | 0.0    | 0.0   | 4.5   | 0.0  | 183.3  | 0.0 | 0.0 | 3.7  | 0.0   | 0.0    | 1334.7  |
| W8-04    | 0.0    | 0.0   | 3.9   | 0.0  | 300.0  | 0.0 | 0.0 | 0.8  | 0.0   | 0.0    | 1111.8  |
| W9-01    | 0.0    | 0.0   | 10.5  | 0.0  | 26.7   | 1.8 | 0.0 | 27.3 | 0.0   | 1537.7 | 1364.0  |
| W9-02    | 0.0    | 0.0   | 23.6  | 0.0  | 350.0  | 2.4 | 0.0 | 22.8 | 0.0   | 1439.3 | 1394.8  |
| W9-03    | 0.0    | 0.0   | 11.6  | 0.0  | 553.3  | 3.7 | 0.0 | 4.2  | 0.0   | 2080.8 | 1916.0  |
| W9-04    | 0.0    | 0.0   | 11.0  | 0.0  | 563.3  | 1.8 | 0.0 | 8.4  | 0.0   | 1759.1 | 1553.2  |
| W10-01   | 0.0    | 0.0   | 35.0  | 0.0  | 500.0  | 0.0 | 0.0 | 17.2 | 89.2  | 0.0    | 419.1   |
| W10-02   | 3.3    | 0.0   | 47.3  | 0.0  | 503.3  | 0.0 | 0.0 | 9.5  | 53.7  | 0.0    | 427.7   |
| W10-03   | 4782.7 | 0.0   | 13.8  | 0.0  | 151.7  | 0.0 | 3.0 | 6.4  | 225.8 | 0.0    | 1489.7  |
| W10-04   | 3.3    | 0.0   | 25.5  | 0.0  | 645.0  | 0.0 | 0.0 | 0.1  | 186.5 | 0.0    | 1024.8  |
| Avg      | 580.3  | 8.2   | 19.9  | 0.9  | 407.6  | 0.3 | 0.1 | 4.2  | 137.0 | 270.4  | 2704.8  |

to provide also strategies for handling roster disruption and rescheduling.

- Finally, our experiments highlight that some wards are undersized in terms of staff available, such that it is not possible to obtain solutions without violations of the Understaffing constraints (A1). This is an issue that can be solved only if managers integrate strategic decisions about workforce planning related to the optimal size of personnel and mix of their skills to meet coverage requirements.

### CRedit authorship contribution statement

**Sara Ceschia:** Conceptualization, Methodology, Software, Writing. **Luca Di Gaspero:** Software, Data curation, Writing, Visualization. **Vincenzo Mazzaracchio:** Conceptualization, Validation, Data curation, Funding. **Giuseppe Policante:** Conceptualization, Validation, Data curation, Funding. **Andrea Schaerf:** Conceptualization, Methodology, Formal analysis, Writing, Supervision.

### Data availability

All data can be downloaded from <https://bitbucket.org/satt/nrp-instances>

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### Appendix. System infrastructure

As reported by Van den Bergh et al. [1], most of the personnel scheduling research projects are not implemented in practice due to the difficult integration of the proposed algorithms into the existing software systems of healthcare institutions.

Therefore, we believe that describing the actual system infrastructure is a noteworthy part of the contribution. Fig. A.1 shows the components of the system architecture that are related to the integration with the solver.

In particular, for the purpose of decoupling the solver from the data management part there are two main architectural blocks: one related to the UI/control and one to the solution process. In principle there are multiple instances of the User Interface, one for each hospital ward but all of them share a common database in which the specific ward data are stored. Also there might be multiple installations of the whole data management system for different hospitals. Conversely, the solver service will be available in a cloud-like manner as an endpoint serving multiple system instances.

An optimization request can be issued from the interface (step ①), which is handled in a task queue that acts as the interface between the UI and the solver part.

Each optimization request contains all the specific data related to the roster to be optimized that is read from the database. The request is selected for execution by a task scheduler (step ②) that performs all the necessary steps for running the solver and returning the solution. In particular, the data coming from

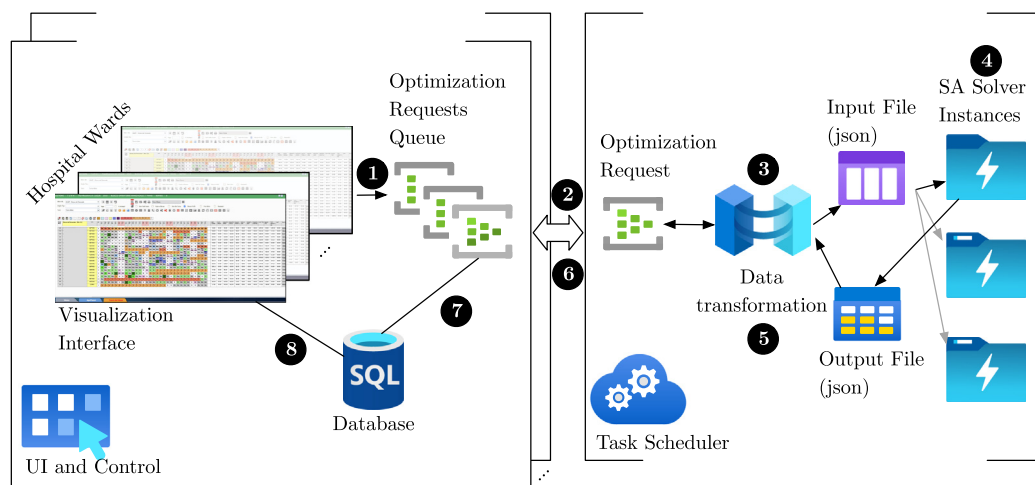


Fig. A.1. System architecture.

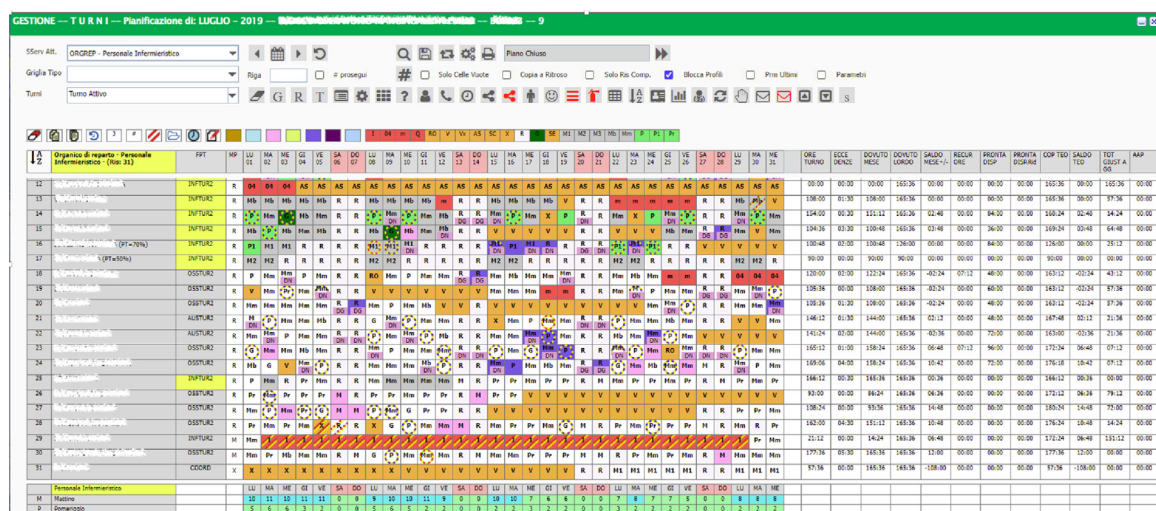


Fig. A.2. An example of a monthly plan for a ward.

the request undergoes a data transformation (step 3) into the JSON input format so that it can be directly handled by the solver (step 4). Since each optimization request is independent from the others, there might be multiple instances of the solver that run concurrently (i.e., in parallel) each handling a different request. Once the solution process has finished another JSON content is created by the solver in the output format, which also undergoes data transformation (step 5). The solution suitably formatted is then returned and written in the database (steps 6 and 7) so that is available for visualization (step 8) and further manipulation on the UI/Control side.

Fig. A.2 shows the UI of the system with an example of a plan of a ward for one month, which is the final product of the solver. The interface allows, for a trained people, an immediate visual inspection of the different solution aspects, in particular presenting also key performance indicators that can be used for validating the solution.

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